

Design specification

AquaElevate

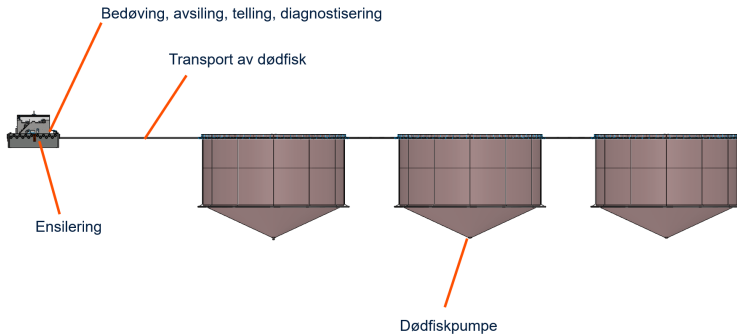
Dead fish handling

Operational systems - ScaleAQ

January 28, 2026



Dødfiskhåndtering fra merd til ensilasje



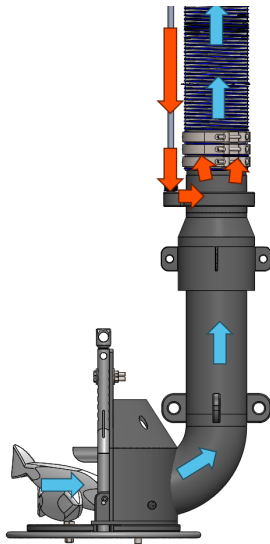
System overview

1. Dead fish are gathered at the bottom of the pen.



System overview

1. Dead fish are gathered at the bottom of the pen.
2. The fish flows through a GRP pipe from the pen and up to the LiftUp unit.



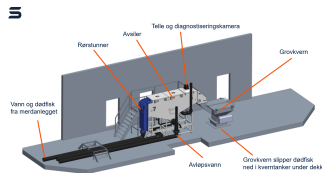
System overview

1. Dead fish are gathered at the bottom of the pen.
2. The fish flows through a GRP pipe from the pen and up to the LiftUp unit.
3. Stunner unit (state mandatory)



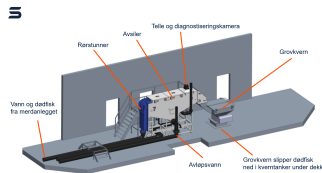
System overview

1. Dead fish are gathered at the bottom of the pen.
2. The fish flows through a GRP pipe from the pen and up to the LiftUp unit.
3. Stunner unit (state mandatory)
4. It then moves through the liftup-unit by means of gravity and a VSD controlled conveyor belt



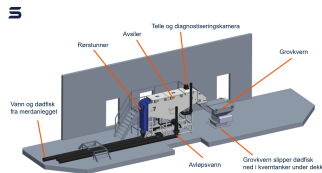
System overview

1. Dead fish are gathered at the bottom of the pen.
2. The fish flows through a GRP pipe from the pen and up to the LiftUp unit.
3. Stunner unit (state mandatory)
4. It then moves through the liftup-unit by means of gravity and a VSD controlled conveyor belt
5. Flushing system



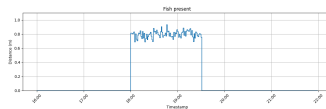
System overview

1. Dead fish are gathered at the bottom of the pen.
2. The fish flows through a GRP pipe from the pen and up to the LiftUp unit.
3. Stunner unit (state mandatory)
4. It then moves through the liftup-unit by means of gravity and a VSD controlled conveyor belt
5. Flushing system
6. Camera unit / vision system



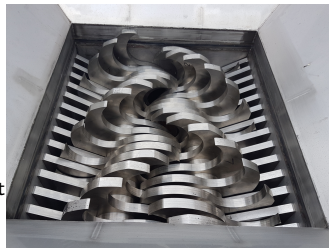
System overview

1. Dead fish are gathered at the bottom of the pen.
2. The fish flows through a GRP pipe from the pen and up to the LiftUp unit.
3. Stunner unit (state mandatory)
4. It then moves through the liftup-unit by means of gravity and a VSD controlled conveyor belt
5. Flushing system
6. Camera unit / vision system
7. Distance sensor



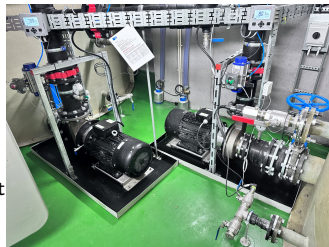
System overview

1. Dead fish are gathered at the bottom of the pen.
2. The fish flows through a GRP pipe from the pen and up to the LiftUp unit.
3. Stunner unit (state mandatory)
4. It then moves through the liftup-unit by means of gravity and a VSD controlled conveyor belt
5. Flushing system
6. Camera unit / vision system
7. Distance sensor
8. Moves through a pre-grinder



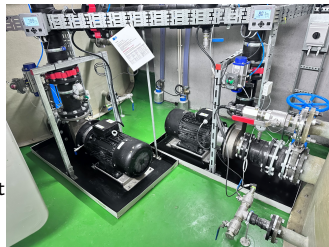
System overview

1. Dead fish are gathered at the bottom of the pen.
2. The fish flows through a GRP pipe from the pen and up to the LiftUp unit.
3. Stunner unit (state mandatory)
4. It then moves through the liftup-unit by means of gravity and a VSD controlled conveyor belt
5. Flushing system
6. Camera unit / vision system
7. Distance sensor
8. Moves through a pre-grinder
9. Stored in tanks and continuously circulated/grinded in the ensilage system.



System overview

1. Dead fish are gathered at the bottom of the pen.
2. The fish flows through a GRP pipe from the pen and up to the LiftUp unit.
3. Stunner unit (state mandatory)
4. It then moves through the liftup-unit by means of gravity and a VSD controlled conveyor belt
5. Flushing system
6. Camera unit / vision system
7. Distance sensor
8. Moves through a pre-grinder
9. Stored in tanks and continuously circulated/grinded in the ensilage system.
10. Automatic handling of acid dosing



Compressor

1. Will be continuously running in subsea applications. Will only run when dead fish system is active on other pens.
4. Has its own local user interface with configuration and setting possibilities.



Compressor

1. Will be continuously running in subsea applications. Will only run when dead fish system is active on other pens.
2. Responsibilities are delivering air to
4. Has its own local user interface with configuration and setting possibilities.



Compressor

1. Will be continuously running in subsea applications. Will only run when dead fish system is active on other pens.
2. Responsibilities are delivering air to
 - ▶ a buffer tank used for refilling the air domes in the subsea pens.
4. Has its own local user interface with configuration and setting possibilities.



Compressor

1. Will be continuously running in subsea applications. Will only run when dead fish system is active on other pens.
2. Responsibilities are delivering air to
 - ▶ a buffer tank used for refilling the air domes in the subsea pens.
 - ▶ the air pump at the bottom of the pens for collecting dead fish.
4. Has its own local user interface with configuration and setting possibilities.



Compressor

1. Will be continuously running in subsea applications. Will only run when dead fish system is active on other pens.
2. Responsibilities are delivering air to
 - ▶ a buffer tank used for refilling the air domes in the subsea pens.
 - ▶ the air pump at the bottom of the pens for collecting dead fish.
3. Controlled and monitored in the PLC by hardwired (+24VDC) signals.
4. Has its own local user interface with configuration and setting possibilities.



LiftUp unit

1. Consists of different subsystems.



LiftUp unit

1. Consists of different subsystems.

- ▶ Stunner



LiftUp unit

1. Consists of different subsystems.

- ▶ Stunner
- ▶ Flusher



LiftUp unit

1. Consists of different subsystems.

- ▶ Stunner
- ▶ Flusher
- ▶ Pen selector control



LiftUp unit

1. Consists of different subsystems.

- ▶ Stunner
- ▶ Flusher
- ▶ Pen selector control
- ▶ Conveyor belt



LiftUp unit

1. Consists of different subsystems.
 - ▶ Stunner
 - ▶ Flusher
 - ▶ Pen selector control
 - ▶ Conveyor belt
 - ▶ Camera diagnostic unit



LiftUp unit

1. Consists of different subsystems.
 - ▶ Stunner
 - ▶ Flusher
 - ▶ Pen selector control
 - ▶ Conveyor belt
 - ▶ Camera diagnostic unit
2. All components connected to Framo's local PLC on the unit.



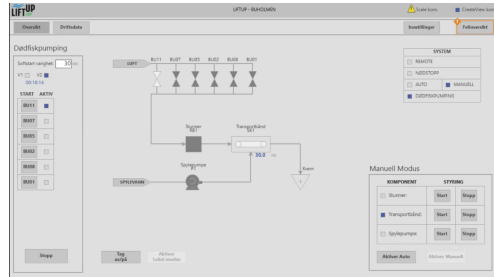
LiftUp unit

1. Consists of different subsystems.
 - ▶ Stunner
 - ▶ Flusher
 - ▶ Pen selector control
 - ▶ Conveyor belt
 - ▶ Camera diagnostic unit
2. All components connected to Framo's local PLC on the unit.
3. Modbus TCP/IP interface to the local PLC for monitoring and control.



LiftUp unit

1. Consists of different subsystems.
 - ▶ Stunner
 - ▶ Flusher
 - ▶ Pen selector control
 - ▶ Conveyor belt
 - ▶ Camera diagnostic unit
2. All components connected to Framo's local PLC on the unit.
3. Modbus TCP/IP interface to the local PLC for monitoring and control.
4. Local user interface



Ensilage unit

1. Consists of different subsystems.



Ensilage unit

1. Consists of different subsystems.



Ensilage unit

1. Consists of different subsystems.



Ensilage unit

1. Consists of different subsystems.



Ensilage unit

1. Consists of different subsystems.



Ensilage unit

1. Consists of different subsystems.



Ensilage unit

1. Consists of different subsystems.

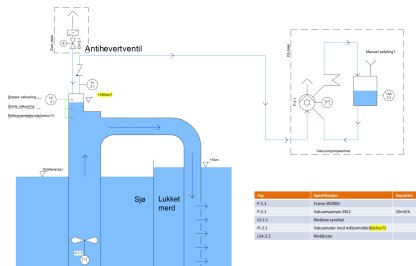


Ensilage unit

1. Consists of different subsystems.



Scenarios

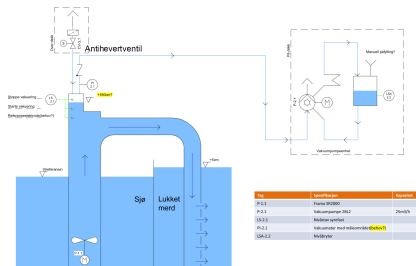


1. All systems are powered up and outlet valve is in position within valid range.

Finish

Scenario 1 - Cold start of inlet pump no. 1 into manual operation mode at a given speed.

Scenarios

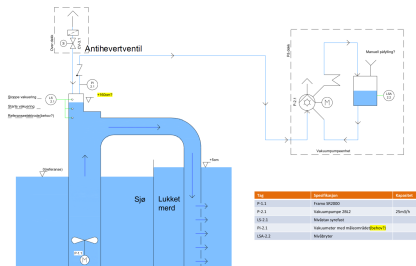


1. All systems are powered up and outlet valve is in position within valid range.
2. User sends start command to pump no. 1

Finish

Scenario 1 - Cold start of inlet pump no. 1 into manual operation mode at a given speed.

Scenarios

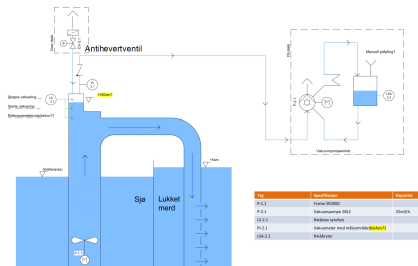


1. All systems are powered up and outlet valve is in position within valid range.
2. User sends start command to pump no. 1
3. System checks condition of process and begins the starting sequence of inlet pump.

Finish

Scenario 1 - Cold start of inlet pump no. 1 into manual operation mode at a given speed.

Scenarios

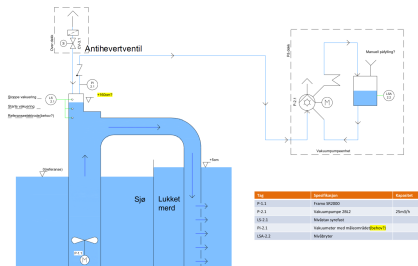


1. All systems are powered up and outlet valve is in position within valid range.
2. User sends start command to pump no. 1
3. System checks condition of process and begins the starting sequence of inlet pump.
4. The pumps starts at reduced speed, vacuum system starts and anti-siphon valve closes.

Finish

Scenario 1 - Cold start of inlet pump no. 1 into manual operation mode at a given speed.

Scenarios

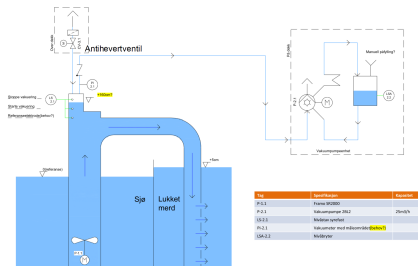


1. All systems are powered up and outlet valve is in position within valid range.
2. User sends start command to pump no. 1
3. System checks condition of process and begins the starting sequence of inlet pump.
4. The pumps starts at reduced speed, vacuum system starts and anti-siphon valve closes.
5. After a limit switch at outlet piping triggers, the vacuum pump stops and the pump starts running at reduced speed

Finish

Scenario 1 - Cold start of inlet pump no. 1 into manual operation mode at a given speed.

Scenarios

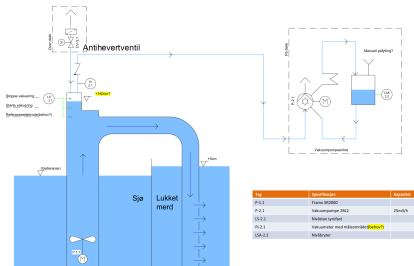


1. All systems are powered up and outlet valve is in position within valid range.
2. User sends start command to pump no. 1
3. System checks condition of process and begins the starting sequence of inlet pump.
4. The pumps starts at reduced speed, vacuum system starts and anti-siphon valve closes.
5. After a limit switch at outlet piping triggers, the vacuum pump stops and the pump starts running at reduced speed
6. System changes to "remote operation" and in "manual mode" at 50 RPM.

Finish

Scenario 1 - Cold start of inlet pump no. 1 into manual operation mode at a given speed.

Scenarios

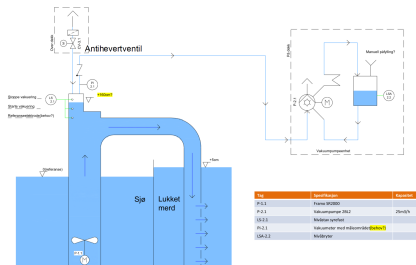


1. All systems are powered up and outlet valve is in position within valid range.

Finish

Scenario 2 - Cold start of inlet pump no. 1 into failed to start state.

Scenarios

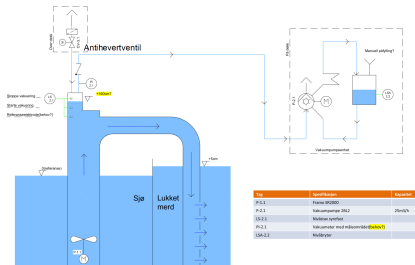


1. All systems are powered up and outlet valve is in position within valid range.
2. User sends start command to pump no. 1

Finish

Scenario 2 - Cold start of inlet pump no. 1 into failed to start state.

Scenarios

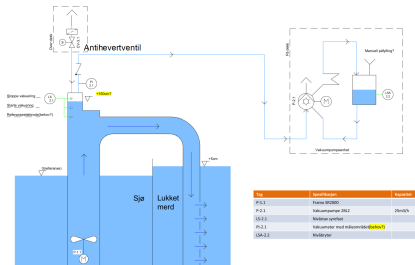


1. All systems are powered up and outlet valve is in position within valid range.
2. User sends start command to pump no. 1
3. System checks condition of process and begins the starting sequence of inlet pump.

Finish

Scenario 2 - Cold start of inlet pump no. 1 into failed to start state.

Scenarios

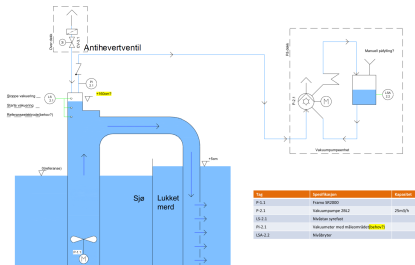


1. All systems are powered up and outlet valve is in position within valid range.
2. User sends start command to pump no. 1
3. System checks condition of process and begins the starting sequence of inlet pump.
4. The pumps starts at reduced speed, vacuum system starts and anti-siphon valve closes.

Finish

Scenario 2 - Cold start of inlet pump no. 1 into failed to start state.

Scenarios

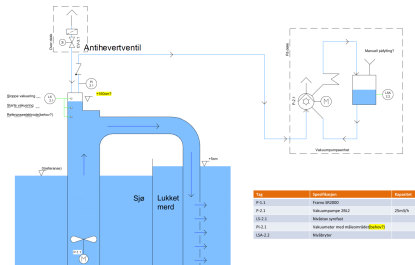


1. All systems are powered up and outlet valve is in position within valid range.
2. User sends start command to pump no. 1
3. System checks condition of process and begins the starting sequence of inlet pump.
4. The pumps starts at reduced speed, vacuum system starts and anti-siphon valve closes.
5. Limit switch has not triggered within the defined time.

Finish

Scenario 2 - Cold start of inlet pump no. 1 into failed to start state.

Scenarios



1. All systems are powered up and outlet valve is in position within valid range.
2. User sends start command to pump no. 1
3. System checks condition of process and begins the starting sequence of inlet pump.
4. The pumps starts at reduced speed, vacuum system starts and anti-siphon valve closes.
5. Limit switch has not triggered within the defined time.
6. System changes to "failed to start", stop command is sent to the pump and anti siphon valve opens.

Finish

Scenario 2 - Cold start of inlet pump no. 1 into failed to start state.

Scenarios

1. All pumps have started and is running in remote operation at 50RPM.

Finish

Scenario 3 - Putting the pumps in auto mode after startup and update the set point to the regulator.

Scenarios

1. All pumps have started and is running in remote operation at 50RPM.
2. Send auto command to controller. Verify that the controller set point is updated with the current value.

Finish

Scenario 3 - Putting the pumps in auto mode after startup and update the set point to the regulator.

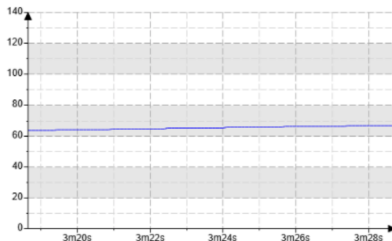
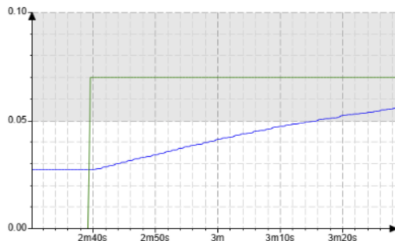
Scenarios

1. All pumps have started and is running in remote operation at 50RPM.
2. Send auto command to controller. Verify that the controller set point is updated with the current value.
3. Verify that the RPM control signal to the pumps are stable at around 50RPM.

Finish

Scenario 3 - Putting the pumps in auto mode after startup and update the set point to the regulator.

Scenarios

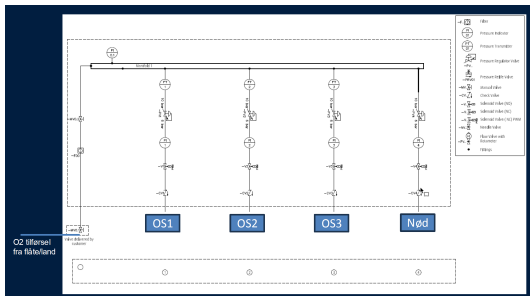


1. All pumps have started and is running in remote operation at 50RPM.
2. Send auto command to controller. Verify that the controller set point is updated with the current value.
3. Verify that the RPM control signal to the pumps are stable at around 50RPM.
4. Change set point to 7cm. Verify that RPM control signal and delta height is stable and rising towards stationary values.

Finish

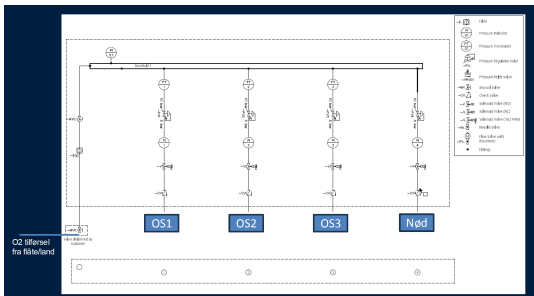
Scenario 3 - Putting the pumps in auto mode after startup and update the set point to the regulator.

Scenarios



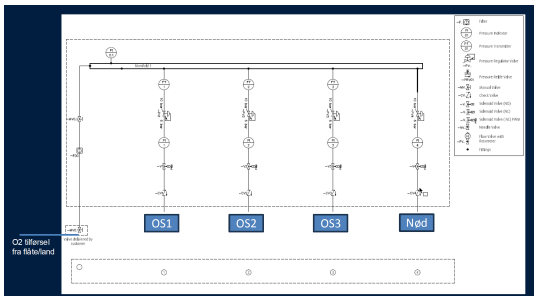
1. Inlet pump no. 1 is running at above minimum speed.

Scenarios



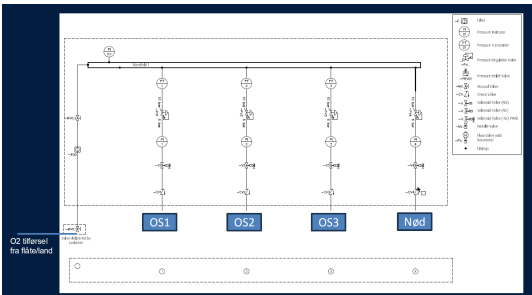
1. Inlet pump no. 1 is running at above minimum speed.
2. We have minimum pressure at oxygen piping manifold.

Scenarios



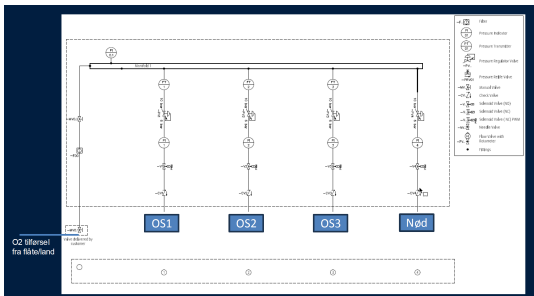
1. Inlet pump no. 1 is running at above minimum speed.
2. We have minimum pressure at oxygen piping manifold.
3. Operator sends start signal to the oxygenator system no. 1.

Scenarios



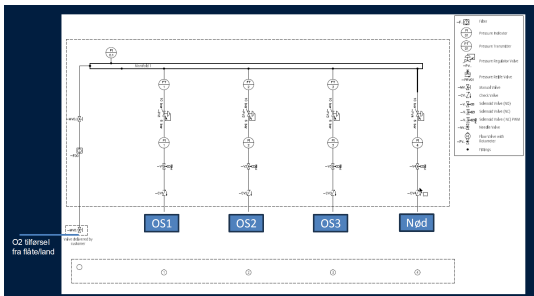
1. Inlet pump no. 1 is running at above minimum speed.
2. We have minimum pressure at oxygen piping manifold.
3. Operator sends start signal to the oxygenator system no. 1.
4. Control valve is set to a manual predefined position.

Scenarios



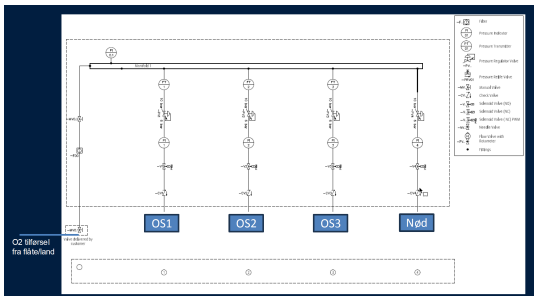
1. Inlet pump no. 1 is running at above minimum speed.
2. We have minimum pressure at oxygen piping manifold.
3. Operator sends start signal to the oxygenator system no. 1.
4. Control valve is set to a manual predefined position.
5. Solenoid valve opens.

Scenarios



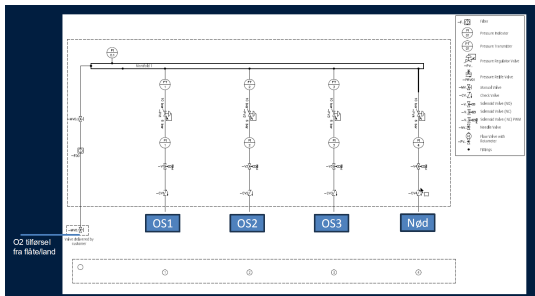
1. Inlet pump no. 1 is running at above minimum speed.
2. We have minimum pressure at oxygen piping manifold.
3. Operator sends start signal to the oxygenator system no. 1.
4. Control valve is set to a manual predefined position.
5. Solenoid valve opens.
6. Operator sends "auto mode" signal.

Scenarios



1. Inlet pump no. 1 is running at above minimum speed.
2. We have minimum pressure at oxygen piping manifold.
3. Operator sends start signal to the oxygenator system no. 1.
4. Control valve is set to a manual predefined position.
5. Solenoid valve opens.
6. Operator sends "auto mode" signal.
7. Operator changes set point to the desired oxygen concentration.

Scenarios



1. Inlet pump no. 1 is running at above minimum speed.
2. We have minimum pressure at oxygen piping manifold.
3. Operator sends start signal to the oxygenator system no. 1.
4. Control valve is set to a manual predefined position.
5. Solenoid valve opens.
6. Operator sends "auto mode" signal.
7. Operator changes set point to the desired oxygen concentration.
8. Operator verifies that the valve control signal and oxygen concentration levels are stable and moving towards a stationary value.